

# Supplementary Material: Leakage-Controlled Concept Graph Construction and Cold-Start Diagnostic Protocol for Knowledge Tracing

Dao Minh Tuan<sup>1\*</sup>, Nguyen Tien Duong<sup>1</sup>, Nguyen Khanh Trinh<sup>1</sup>,  
Nguyen Van Hau<sup>1</sup>, Le Hoang Son<sup>2</sup>

<sup>1\*</sup>Hung Yen University of Technology and Education, Hung Yen,  
Vietnam.

<sup>2</sup>Information Technology Institute, Vietnam National University, 144  
Xuan Thuy Street, Cau Giay, Hanoi, Vietnam.

## Abstract

This supplement provides per-dataset baseline and ablation tables, paired significance tests, full cold-start strata, DDR downstream scatter plot, sequence-autocorrelation statistics, operator definitions, ground-truth disagreement examples, extended ground-truth metrics with node Jaccard, and multi-benchmark schematic figures omitted from the main manuscript for length.

## Supplementary index

**Tables.** S1–S5: per-dataset baseline diagnostics (Section 1); S6–S10: per-dataset graph ablation (Section 2); S11: paired significance tests on public benchmarks (Section 3); S12–S13: cold-start diagnostics (Section 8); S14: ground-truth metrics including node Jaccard (Section 7).

**Sections and figures.** Section S3: sequence-autocorrelation analysis (Section 4); Section S4: DDR augmentation operators (Section 5); Section S5: ground-truth disagreement examples (Section 6); Figure S1: multi-benchmark KT-graph schematics (Figure 10); Figure S2: multi-benchmark DDR curves (Figure 10); Figure S3: DDR downstream scatter plot (Figure 9).

**Table 1:** Diagnostic baseline results (train-only graphs). Baselines are used for protocol comparison only; no SOTA claim is made. Values are fold means with 95% bootstrap confidence intervals over valid+test sequence positions (`n_folds` as exported per dataset).

| Model           | AUC   |                | ACC   |                | NLL   |                |
|-----------------|-------|----------------|-------|----------------|-------|----------------|
| AKT             | 0.984 | [0.982, 0.989] | 0.951 | [0.945, 0.962] | 0.121 | [0.093, 0.135] |
| BKT             | 0.780 | [0.779, 0.781] | 0.846 | [0.845, 0.847] | 0.379 | [0.377, 0.380] |
| DGEKT           | 0.981 | [0.981, 0.981] | 0.943 | [0.942, 0.943] | 0.137 | [0.136, 0.138] |
| DKT             | 0.984 | [0.982, 0.989] | 0.951 | [0.945, 0.962] | 0.120 | [0.093, 0.134] |
| DyGKT           | 0.978 | [0.978, 0.978] | 0.934 | [0.933, 0.934] | 0.151 | [0.151, 0.152] |
| GIKT            | 0.982 | [0.982, 0.982] | 0.946 | [0.946, 0.946] | 0.133 | [0.132, 0.133] |
| <i>simpleKT</i> | 0.984 | [0.982, 0.989] | 0.951 | [0.945, 0.962] | 0.121 | [0.093, 0.135] |
| SKT             | 0.979 | [0.976, 0.986] | 0.938 | [0.931, 0.954] | 0.139 | [0.107, 0.156] |

**Table 2:** Diagnostic baselines: ASSISTments 2012 (three-fold learner CV (`n_folds=3`); train-only graphs).

| Model           | AUC   |                | ACC   |                | NLL   |                |
|-----------------|-------|----------------|-------|----------------|-------|----------------|
| AKT             | 0.968 | [0.968, 0.969] | 0.916 | [0.914, 0.917] | 0.175 | [0.173, 0.176] |
| BKT             | 0.669 | [0.667, 0.671] | 0.713 | [0.711, 0.715] | 0.574 | [0.573, 0.576] |
| DGEKT           | 0.960 | [0.959, 0.961] | 0.907 | [0.906, 0.908] | 0.186 | [0.185, 0.188] |
| DKT             | 0.963 | [0.961, 0.964] | 0.910 | [0.908, 0.911] | 0.182 | [0.180, 0.185] |
| DyGKT           | 0.952 | [0.951, 0.952] | 0.899 | [0.898, 0.900] | 0.200 | [0.198, 0.203] |
| GIKT            | 0.967 | [0.967, 0.968] | 0.915 | [0.914, 0.915] | 0.175 | [0.174, 0.177] |
| GKT             | 0.961 | [0.960, 0.962] | 0.907 | [0.906, 0.908] | 0.185 | [0.184, 0.187] |
| <i>simpleKT</i> | 0.968 | [0.967, 0.968] | 0.916 | [0.915, 0.916] | 0.173 | [0.172, 0.175] |
| SKT             | 0.953 | [0.952, 0.953] | 0.898 | [0.897, 0.899] | 0.198 | [0.196, 0.200] |

## 1 Per-dataset baseline diagnostics

Tables S1–S5 (Tables 1–5) reproduce the five diagnostic baseline tables (AUC, ACC, NLL with 95% bootstrap CIs) that the main manuscript summarises in its cross-dataset AUC summary table.

## 2 Per-dataset graph construction ablation

Tables S6–S10 (Table 6–10) report fold 0 train-only versus full-log metrics for each graph-augmented pyKT model. The main manuscript reports only the maximum  $|\Delta\text{AUC}|$  and  $|\Delta\text{ACC}|$  per dataset.

## 3 Paired significance tests (public benchmarks)

Table S11 (Table 11) reports paired  $t$ -test and Wilcoxon  $p$ -values for graph-augmented models versus *simpleKT* on the three public benchmarks. The main manuscript baseline-CV table lists only Wilcoxon  $p$ -values because three-fold CV limits Wilcoxon resolution ( $p_{\min}=0.25$ ).

**Table 3:** Diagnostic baselines: XES3G5M (three-fold learner CV (`n_folds=3`); train-only graphs).

| Model           | AUC   |                | ACC   |                | NLL   |                |
|-----------------|-------|----------------|-------|----------------|-------|----------------|
| AKT             | 0.875 | [0.874, 0.875] | 0.858 | [0.857, 0.858] | 0.324 | [0.323, 0.326] |
| BKT             | 0.636 | [0.634, 0.638] | 0.793 | [0.791, 0.793] | 0.490 | [0.490, 0.492] |
| DGEKT           | 0.841 | [0.841, 0.842] | 0.844 | [0.844, 0.845] | 0.358 | [0.356, 0.358] |
| DKT             | 0.858 | [0.857, 0.858] | 0.851 | [0.851, 0.852] | 0.342 | [0.341, 0.342] |
| DyGKT           | 0.752 | [0.752, 0.754] | 0.813 | [0.813, 0.814] | 0.431 | [0.429, 0.431] |
| GIKT            | 0.878 | [0.878, 0.878] | 0.860 | [0.859, 0.860] | 0.321 | [0.320, 0.322] |
| GKT             | 0.834 | [0.833, 0.835] | 0.842 | [0.841, 0.842] | 0.364 | [0.363, 0.365] |
| <i>simpleKT</i> | 0.875 | [0.874, 0.875] | 0.858 | [0.857, 0.859] | 0.324 | [0.323, 0.325] |
| SKT             | 0.625 | [0.623, 0.626] | 0.800 | [0.799, 0.801] | 0.482 | [0.480, 0.483] |

**Table 4:** Diagnostic baselines: Synthetic C2 (three-fold learner CV (`n_folds=3`); train-only graphs).

| Model           | AUC   |                | ACC   |                | NLL   |                |
|-----------------|-------|----------------|-------|----------------|-------|----------------|
| AKT             | 0.873 | [0.870, 0.876] | 0.808 | [0.802, 0.812] | 0.407 | [0.401, 0.417] |
| BKT             | 0.835 | [0.833, 0.838] | 0.782 | [0.780, 0.784] | 0.460 | [0.458, 0.462] |
| DGEKT           | 0.710 | [0.634, 0.859] | 0.724 | [0.686, 0.794] | 0.551 | [0.432, 0.613] |
| DKT             | 0.862 | [0.857, 0.868] | 0.798 | [0.788, 0.805] | 0.421 | [0.413, 0.434] |
| DyGKT           | 0.710 | [0.631, 0.867] | 0.726 | [0.686, 0.801] | 0.550 | [0.421, 0.617] |
| GIKT            | 0.875 | [0.871, 0.880] | 0.809 | [0.802, 0.815] | 0.405 | [0.398, 0.416] |
| GKT             | 0.873 | [0.868, 0.878] | 0.808 | [0.799, 0.814] | 0.407 | [0.400, 0.420] |
| <i>simpleKT</i> | 0.875 | [0.871, 0.880] | 0.809 | [0.803, 0.814] | 0.405 | [0.399, 0.416] |
| SKT             | 0.708 | [0.634, 0.853] | 0.722 | [0.686, 0.788] | 0.554 | [0.440, 0.613] |

## 4 Sequence autocorrelation analysis

For every learner we order interactions by timestamp and, over consecutive within-learner pairs  $(t-1, t)$ , measure KC-repeat rate, item-repeat rate, label persistence, lag-one label autocorrelation  $\rho$ , and the AUC of the trivial previous-answer predictor (`scripts/compute_autocorrelation.py`).

## 5 DDR augmentation operator definitions

All operators act on the retained acyclic  $E_{\text{pre}}$  at perturbation strength  $p \in \{0.05, 0.10, 0.20, 0.30\}$ :

- **Attribute masking** masks a random subset of edge attributes without changing  $E_{\text{pre}}$  (DDR=0 by construction).
- **Edge dropping** removes each edge independently with probability  $p$  (expected DDR $\approx p$ ).
- **Subgraph sampling** performs a random walk until  $\lceil (1-p)|\mathcal{C}| \rceil$  nodes are visited and returns the induced subgraph (boundary edges drop; internal connectivity preserved).

**Table 5:** Diagnostic baselines: Synthetic C5 (three-fold learner CV (`n_folds=3`); train-only graphs).

| Model           | AUC   |                | ACC   |                | NLL   |                |
|-----------------|-------|----------------|-------|----------------|-------|----------------|
| AKT             | 0.822 | [0.819, 0.823] | 0.750 | [0.746, 0.752] | 0.500 | [0.496, 0.507] |
| BKT             | 0.791 | [0.789, 0.794] | 0.724 | [0.722, 0.726] | 0.534 | [0.532, 0.536] |
| DGEKT           | 0.830 | [0.827, 0.832] | 0.756 | [0.755, 0.757] | 0.489 | [0.486, 0.492] |
| DKT             | 0.826 | [0.825, 0.827] | 0.754 | [0.752, 0.756] | 0.493 | [0.491, 0.496] |
| DyGKT           | 0.826 | [0.824, 0.828] | 0.751 | [0.748, 0.753] | 0.492 | [0.490, 0.496] |
| GIKT            | 0.820 | [0.817, 0.824] | 0.749 | [0.744, 0.754] | 0.500 | [0.496, 0.504] |
| GKT             | 0.810 | [0.806, 0.814] | 0.738 | [0.730, 0.744] | 0.526 | [0.513, 0.538] |
| <i>simpleKT</i> | 0.829 | [0.828, 0.831] | 0.755 | [0.754, 0.757] | 0.491 | [0.489, 0.492] |
| SKT             | 0.810 | [0.808, 0.813] | 0.740 | [0.737, 0.744] | 0.511 | [0.507, 0.513] |

**Table 6:** Ablation contrasting *train-only* graph smoothing (protocol) with a *full-log* graph built from all learners before splitting (leaky construction). Graph-augmented rows are pyKT checkpoints consuming exported adjacency from P0.  $\Delta$  is fold 0 full-log minus train-only. *to*: train-only graph; *fl*: full-log graph.

| Model | AUC<br>to | AUC<br>fl | $\Delta$ AUC | ACC<br>to | ACC<br>fl | $\Delta$ ACC |
|-------|-----------|-----------|--------------|-----------|-----------|--------------|
| GIKT  | 0.982     | 0.982     | 0.000        | 0.946     | 0.946     | -0.000       |
| SKT   | 0.979     | 0.976     | -0.003       | 0.938     | 0.931     | -0.008       |
| DyGKT | 0.978     | 0.978     | -0.000       | 0.934     | 0.934     | 0.000        |
| DGEKT | 0.981     | 0.981     | -0.000       | 0.943     | 0.943     | -0.000       |

**Table 7:** Train-only vs. full-log graph ablation: ASSISTments 2012.

| Model | AUC<br>to | AUC<br>fl | $\Delta$ AUC | ACC<br>to | ACC<br>fl | $\Delta$ ACC |
|-------|-----------|-----------|--------------|-----------|-----------|--------------|
| GKT   | 0.961     | 0.961     | -0.000       | 0.907     | 0.907     | -0.000       |
| GIKT  | 0.967     | 0.968     | 0.000        | 0.915     | 0.915     | 0.000        |
| SKT   | 0.953     | 0.953     | 0.000        | 0.898     | 0.898     | 0.000        |
| DyGKT | 0.952     | 0.952     | -0.000       | 0.899     | 0.899     | -0.000       |
| DGEKT | 0.960     | 0.960     | 0.000        | 0.907     | 0.907     | 0.000        |

- **Node dropping** removes  $[p|\mathcal{C}|]$  nodes and all incident edges.
- **prereq-preserve** draws the same per-edge budget as edge dropping but removes only transitively redundant edges, protecting the transitive-reduction backbone.

## 6 Ground-truth disagreement examples (Junyi)

The cross-validation procedure exports qualitative disagreement cases to `results/tables/disagreement_examples.csv`. Inferred-only edges are predominantly bidirectional co-occurring arithmetic pairs (e.g. `number_line`  $\leftrightarrow$  `addition_1`). Expert-only edges are systematic level-progression annotations (e.g. `equivalent_fractions_2`  $\rightarrow$

**Table 8:** Train-only vs. full-log graph ablation: XES3G5M.

| Model | AUC<br>to | AUC<br>fl | $\Delta$ AUC | ACC<br>to | ACC<br>fl | $\Delta$ ACC |
|-------|-----------|-----------|--------------|-----------|-----------|--------------|
| GKT   | 0.834     | 0.835     | 0.002        | 0.842     | 0.842     | 0.000        |
| GIKT  | 0.878     | 0.878     | 0.000        | 0.860     | 0.860     | 0.000        |
| SKT   | 0.625     | 0.624     | -0.000       | 0.800     | 0.800     | 0.000        |
| DyGKT | 0.752     | 0.749     | -0.003       | 0.813     | 0.812     | -0.001       |
| DGEKT | 0.841     | 0.841     | -0.000       | 0.844     | 0.844     | -0.001       |

**Table 9:** Train-only vs. full-log graph ablation: Synthetic C2.

| Model | AUC<br>to | AUC<br>fl | $\Delta$ AUC | ACC<br>to | ACC<br>fl | $\Delta$ ACC |
|-------|-----------|-----------|--------------|-----------|-----------|--------------|
| GKT   | 0.873     | 0.873     | -0.001       | 0.808     | 0.807     | -0.001       |
| GIKT  | 0.875     | 0.874     | -0.000       | 0.809     | 0.809     | 0.000        |
| SKT   | 0.708     | 0.708     | 0.000        | 0.722     | 0.723     | 0.001        |
| DyGKT | 0.710     | 0.710     | 0.000        | 0.726     | 0.726     | -0.000       |
| DGEKT | 0.710     | 0.711     | 0.001        | 0.724     | 0.725     | 0.001        |

**Table 10:** Train-only vs. full-log graph ablation: Synthetic C5.

| Model | AUC<br>to | AUC<br>fl | $\Delta$ AUC | ACC<br>to | ACC<br>fl | $\Delta$ ACC |
|-------|-----------|-----------|--------------|-----------|-----------|--------------|
| GKT   | 0.810     | 0.812     | 0.002        | 0.738     | 0.738     | 0.000        |
| GIKT  | 0.820     | 0.820     | -0.000       | 0.749     | 0.747     | -0.002       |
| SKT   | 0.810     | 0.809     | -0.001       | 0.740     | 0.739     | -0.001       |
| DyGKT | 0.826     | 0.826     | 0.000        | 0.751     | 0.751     | 0.001        |
| DGEKT | 0.830     | 0.830     | 0.000        | 0.756     | 0.756     | -0.001       |

equivalent\_fractions). The `wrong_direction` category contains many reversed level-progression pairs such as `addition_1`  $\rightarrow$  `addition_2`.

## 7 Ground-truth metrics with node Jaccard

Table S14 (Table 13) extends the main manuscript ground-truth table with node Jaccard across the same top- $K$  cutoffs.

## 8 Cold-start tables (full)

Table S12 (Table 14) reports fold 0 *simpleKT* metrics by KC-frequency stratum; Table S13 (Table 15) lists three-fold per-model AUC and  $\Delta_{\max}$  by stratum. The main manuscript summarises both in a single table.

**Table 11:** Paired comparisons against *simpleKT* on public benchmarks (three-fold learner CV).  $\Delta$  is mean AUC difference (model minus *simpleKT*);  $p_t$  is the paired two-sample  $t$ -test  $p$ -value;  $p_W$  is the Wilcoxon signed-rank  $p$ -value (minimum attainable 0.25 with three folds).

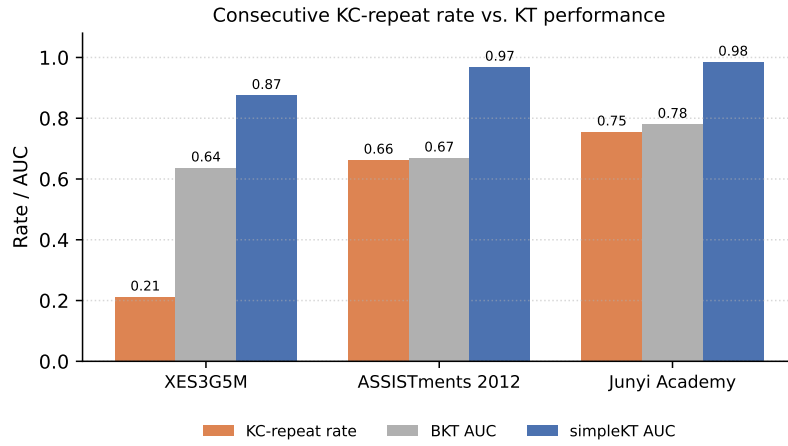
| Dataset          | Model | $\Delta$ AUC | $p_t$  | $p_W$ |
|------------------|-------|--------------|--------|-------|
| ASSISTments 2012 | GKT   | -0.0069      | <0.001 | 0.25  |
|                  | GIKT  | -0.0005      | 0.001  | 0.25  |
|                  | SKT   | -0.0154      | <0.001 | 0.25  |
|                  | DyGKT | -0.0160      | <0.001 | 0.25  |
|                  | DGEKT | -0.0079      | <0.001 | 0.25  |
| XES3G5M          | GKT   | -0.0411      | <0.001 | 0.25  |
|                  | GIKT  | +0.0031      | <0.001 | 0.25  |
|                  | SKT   | -0.2501      | <0.001 | 0.25  |
|                  | DyGKT | -0.1223      | <0.001 | 0.25  |
|                  | DGEKT | -0.0334      | <0.001 | 0.25  |
| Junyi Academy    | GIKT  | -0.0021      | 0.445  | 1.00  |
|                  | SKT   | -0.0055      | 0.031  | 0.25  |
|                  | DyGKT | -0.0066      | 0.100  | 0.25  |
|                  | DGEKT | -0.0034      | 0.267  | 0.25  |

**Table 12:** Sequence-autocorrelation diagnostics on the three public benchmarks. *KC rep.* and *Item rep.* are the fractions of consecutive within-learner interactions that repeat the previous KC or item; *Persist. acc.* is  $P(y_t=y_{t-1})$ ; *Lag-1  $\rho$*  is the lag-one label autocorrelation; *Prev-AUC* is the AUC of the trivial predictor  $\hat{y}_t=y_{t-1}$ .

| Dataset          | KC rep. | Item rep. | Persist. acc. | Lag-1 $\rho$ | Prev-AUC |
|------------------|---------|-----------|---------------|--------------|----------|
| Junyi Academy    | 0.755   | 0.755     | 0.786         | 0.244        | 0.622    |
| ASSISTments 2012 | 0.662   | 0.000     | 0.664         | 0.207        | 0.604    |
| XES3G5M          | 0.211   | 0.145     | 0.786         | 0.342        | 0.671    |

## 9 DDR downstream scatter plot

## 10 Additional schematic figures



**Fig. 1:** Consecutive KC-repeat rate against *simpleKT* AUC across the three public benchmarks.

**Table 13:** Ground-truth cross-validation on Junyi: edge/direction/reachability metrics and node Jaccard across top- $K$  cutoffs. The main manuscript table reports edge precision/recall, direction agreement, and reachability F1 only.

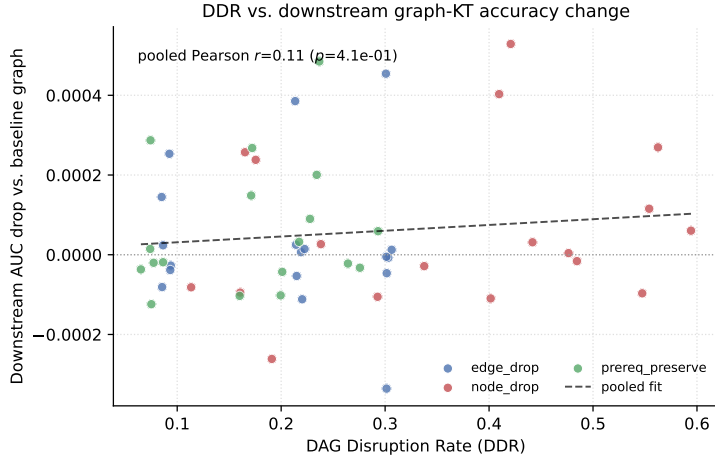
| $K$                    | Edge F1 | Dir. agr. | Rch. F1 | Node Jaccard |
|------------------------|---------|-----------|---------|--------------|
| 200                    | 0.071   | 0.346     | 0.008   | 0.290        |
| 500                    | 0.109   | 0.225     | 0.021   | 0.453        |
| 800                    | 0.150   | 0.280     | 0.062   | 0.572        |
| $ \text{expert} =1131$ | 0.183   | 0.261     | 0.104   | 0.632        |
| 1200                   | 0.185   | 0.262     | 0.126   | 0.641        |
| 1800                   | 0.188   | 0.278     | 0.260   | 0.644        |
| 2500                   | 0.187   | 0.285     | 0.339   | 0.610        |
| 5000 (cap)             | 0.187   | 0.285     | 0.339   | 0.610        |

**Table 14:** Per-stratum cold-start KC diagnostic with *simpleKT* on fold 0. The  $n$  column counts test interactions whose KC falls in the stratum train-frequency bin; *out of range* marks test KCs outside the four canonical bins. Stratum mass varies by dataset split; under default bins Synthetic C2/C5 route all test interactions into the hot stratum ( $> 500$  train-fold hits per knowledge component).

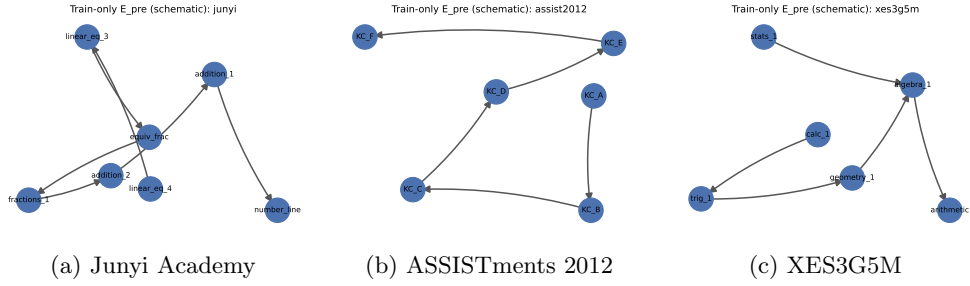
| Dataset       | Stratum      | $n$       | AUC   | ACC   | NLL   |
|---------------|--------------|-----------|-------|-------|-------|
| Junyi Academy | very_cold    | 19        | 1.000 | 0.684 | 0.456 |
|               | cold         | 256       | 0.447 | 0.621 | 0.633 |
|               | warm         | 11,257    | 0.615 | 0.727 | 0.572 |
|               | hot          | 7,725,588 | 0.718 | 0.831 | 0.413 |
|               | out of range | 10        | 0.500 | 0.900 | 0.511 |
| ASSIST'12     | very_cold    | 49        | 0.608 | 0.755 | 0.631 |
|               | cold         | 707       | 0.769 | 0.717 | 0.565 |
|               | warm         | 9,017     | 0.765 | 0.687 | 0.587 |
|               | hot          | 807,178   | 0.693 | 0.704 | 0.574 |
| XES3G5M       | very_cold    | 161       | 0.810 | 0.814 | 0.424 |
|               | cold         | 2,592     | 0.798 | 0.794 | 0.453 |
|               | warm         | 17,655    | 0.790 | 0.805 | 0.431 |
|               | hot          | 1,902,424 | 0.717 | 0.796 | 0.467 |
|               | out of range | 6         | 0.625 | 0.667 | 0.638 |
| Synthetic C2  | hot          | 60,000    | 0.624 | 0.683 | 0.607 |
| Synthetic C5  | hot          | 60,000    | 0.630 | 0.611 | 0.651 |

**Table 15:** Cold-start comparison of sequence-only (No-Graph) and graph-aware (Graph) KT models. Entries are three-fold mean AUC by KC train-frequency stratum;  $\Delta_{\max}$  is the best Graph AUC minus the best No-Graph AUC in the same dataset/stratum.

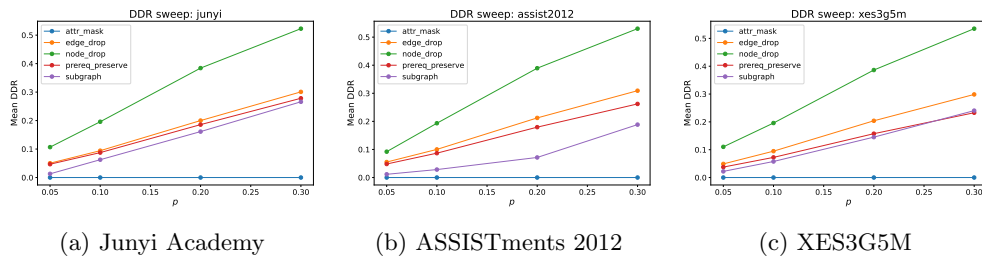
| Dataset      | Stratum   | No-Graph |          | Graph |       |       |                 |
|--------------|-----------|----------|----------|-------|-------|-------|-----------------|
|              |           | DKT      | SimpleKT | GKT   | GIKT  | DGEKT | $\Delta_{\max}$ |
| assist2012   | very_cold | 0.723    | 0.724    | 0.695 | 0.726 | 0.724 | +0.002          |
|              | cold      | 0.769    | 0.772    | 0.750 | 0.772 | 0.772 | +0.000          |
|              | warm      | 0.751    | 0.756    | 0.724 | 0.757 | 0.756 | +0.001          |
|              | hot       | 0.687    | 0.696    | 0.601 | 0.691 | 0.687 | -0.005          |
| junyi        | very_cold | 0.813    | 0.813    | 0.813 | 0.813 | 0.813 | +0.000          |
|              | cold      | 0.505    | 0.505    | 0.505 | 0.505 | 0.505 | +0.000          |
|              | warm      | 0.631    | 0.631    | 0.626 | 0.627 | 0.626 | -0.004          |
|              | hot       | 0.719    | 0.719    | 0.704 | 0.713 | 0.711 | -0.006          |
| synthetic_c2 | hot       | 0.619    | 0.626    | 0.554 | 0.634 | 0.634 | +0.008          |
| synthetic_c5 | hot       | 0.628    | 0.630    | 0.524 | 0.633 | 0.633 | +0.002          |
| xes3g5m      | very_cold | 0.820    | 0.817    | 0.826 | 0.818 | 0.818 | +0.005          |
|              | cold      | 0.802    | 0.804    | 0.793 | 0.804 | 0.803 | -0.000          |
|              | warm      | 0.783    | 0.787    | 0.745 | 0.780 | 0.776 | -0.007          |
|              | hot       | 0.708    | 0.717    | 0.606 | 0.709 | 0.703 | -0.009          |



**Fig. 2:** Supplementary Figure S3. Downstream DGEKT sensitivity to prerequisite-graph disruption on ASSISTments 2012 and XES3G5M. Each point is one fold/operator/strength run; the vertical axis is AUC drop relative to the unperturbed train-only graph.



**Fig. 3:** Supplementary Figure S1. Schematic visualisations of high-support prerequisite chains in train-only  $E_{pre}$  after DAG audit (representative subgraphs). The main manuscript shows the Junyi panel only.



**Fig. 4:** Supplementary Figure S2. DDR versus perturbation strength  $p$  for five augmentation families across three benchmarks. The main manuscript shows the Junyi panel only.